

Neuro-Fusion Graph Networks for Context-Aware Data Reasoning in Heterogeneous Computing Ecosystems

Abstract:

The growing complexity of heterogeneous computing ecosystems—spanning edge devices, cloud infrastructure, and IoT-enabled systems—demands advanced reasoning capabilities that can adapt to dynamic and context-rich environments. This paper introduces **Neuro-Fusion Graph Networks (NFGNs)**, a novel neural architecture that integrates neuro-symbolic learning and graph-based data fusion to enable real-time, context-aware reasoning across diverse data modalities and computing layers. By leveraging adaptive graph attention mechanisms and hierarchical context encoding, NFGNs model relational dependencies between heterogeneous data sources, such as sensor feeds, user behavior patterns, and device-level performance metrics. The proposed framework addresses key challenges in distributed intelligence, including semantic inconsistency, spatiotemporal alignment, and cross-domain interoperability. Extensive experiments on benchmark datasets and real-world edge-cloud environments demonstrate that NFGNs significantly outperform existing models in terms of inference accuracy, context generalization, and system responsiveness. This work lays the groundwork for future intelligent systems that require seamless integration of symbolic reasoning, deep learning, and heterogeneous context modeling.

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